YANG Xuehuan

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Career Objective

- **Key Skills**: AI Security · Autonomous Systems · Sensor Perception · LLM
- **Summary**: Building on experience in sensor perception, vehicle integration, and system intelligence, I am expanding into LLM and voicebot domains to develop multimodal AI systems that bridge autonomous perception with language, enabling safer and more natural human–machine interaction in automotive applications.

Working Experience

- **Singapore Technologies Engineering (ST Engineering IHQ)** Singapore
  Physical AI Engineer, Autonomous System of Systems, Group Engineering Center (GEC) *April 2026 -*
  - **ST AME**: Contributed to a software framework for developing multi-robot systems. Developed SLAM using VIO and LIO for the robot navigation system. Built and integrated the execution and communication layers, and validated system in simulation (Isaac Sim, AirSim) before real-world deployment.
  - **ST Land Systems**: Developed and tested VLA models for autonomous robot navigation. Set up OmniVLA and OpenVLA pipelines, collected real-world robot data, fine-tuned the models, and validated.
  - **ST HTX Humanoid**: Supported the HTX with ST Engineering projects for Home Team Humanoid Robotics Centre (H2RC), which trains advanced humanoid robots with Embodied AI capabilities for frontline safety operations. Performed camera-lidar SLAM for the humanoid navigation system.
- **SingPilot** Singapore
  AI Engineer, Autonomous Perception Team *Feb 2023 - Oct 2025*
  - **PSA EPM**: Participated in deployment and integration of the first XCMG Electric Prime Mover(EPM) at PSA, Singapore. Collaborated with Battery Charging and Swapping System(BCSS) teams to integrate vehicle communication, charging, and hydrogen refueling processes. Experimented with LLM-driven log parsing to automatically extract and interpret fault codes from raw CAN data streams.
  - **PSA HPM**: Participated in the deployment and customer-specific customization of the first XCMG Hydrogen Prime Mover(HPM) at Singapore Port (Demo). Integrated vehicle charging protocols and hydrogen refueling processes in collaboration with NTU teams. Performed CAN bus data acquisition, error code analysis, and on-site troubleshooting to ensure system reliability. Prepared technical reports summarizing results, feedback, and recommendations for performance optimization.
  - **PSA APM**: Worked with perception teams to enhance autonomous driving data models for improved detection accuracy and efficiency. Implemented real-world validation of box and pedestrian detection modules on Terberg prime movers. Authored and maintained technical documentation, ensuring compliance with system integration standards.Collaborated on a pilot voice-assisted testing tool that combined ASR commands with LLM reasoning to streamline on-site validation tasks
  - **NTU COMS**: Collected and analyzed localization and perception data from autonomous electric vehicles in port environments. Developed ROS-based integration and validation workflows on Ubuntu systems, using scripting for automation and troubleshooting. Explored natural-language query interfaces to allow engineers to interact with test data and logs using conversational prompts powered by LLMs.
  - **NTU PSA Electromagnetic**: Integrated autonomous vehicle software with terminal tractors to verify hardware–software compatibility. Conducted electromagnetic field tests to validate sensor stability under operational conditions. Collaborated on early-stage research integrating LLM-powered diagnostic chatbots to assist engineers in identifying and resolving sensor-related issues.
- **School of Computer Science and Engineering@NTU** Singapore
  Research Associate, Cyber Security Team *Feb 2022 - Dec 2022*
  - **AV Lane Attack**: Developed and executed experiments on autonomous vehicle security, focusing on realistic and stealthy adversarial triggers. Processed LiDAR and camera sensor data using Python for robotic system analysis and scenario simulation.
  - **Simulation LGSVL/CARLA Testbed**: Tested autonomous vehicle software on the Apollo simulation platform within the campus, performing robustness tests with pedestrian obstacles and corner cases to validate attack models. Prepared and processed test data to support AV system evaluation and improvement.
- **Centre for Autonomous Robotics Lab, EEE@NTU** Singapore
  Research Intern, Autonomous Vehicle Localization Team *Jan 2019 - Dec 2019*

- **AV Cyber Attack**: Collaborated with Singapore's DSTA on a Ministry of Defense project, focusing on the safety and cybersecurity of driverless cars. Assessed cybersecurity risks across hardware, software, and system-level components. Focused on the safety and cybersecurity of driverless vehicles, addressing risks across hardware, software, and system-level components.
- **GPS Attack**: Conducted GPS signal amplification tests and physical attack experiments to evaluate sensor vulnerabilities. • Processed and classified LiDAR and GPS data with Python, applying algorithmic compensations to enhance system robustness. • Successfully tested autonomous vehicle localization on CETRAN testbed.

• **Centre for Energy Research Institute@NTU** Singapore
Research Intern, Future Mobility Team *Sep 2017 - Dec 2018*

- **1st Driverless 7m Bus NTU-JTC-Navya**: Integrated lane detection results into the vehicle's perception stack for real-time autonomous driving. Applied ZED camera-based lane detection on campus autonomous vehicles, ensuring system reliability and safe navigation.
- **1st Autonomous 12m Bus NTU-Volvo-LTA**: • Contributed to Singapore's first 12-meter autonomous electric bus by developing tools to calibrate sensors without traditional checkerboards. • Collected and processed 3D stereo camera data using CUDA for lane detection and obstacle recognition. • Applied deep learning modules to support real-time autonomous navigation and operational reliability.

Education

• **Nanyang Technological University(NTU)** Singapore
Master of Engineering in Computer Science(SCSE) *Jan 2020 - Dec 2021*

- **Thesis**: Simulation-Based perception testing for autonomous vehicles(10356/154942)
- **Research**: Assisting team with the first physical attack on lane detection systems in autonomous driving, focusing on vulnerabilities and solutions.

• **Nanyang Technological University(NTU)** Singapore
Bachelor of Engineering in Electrical and Electronics (EEE) *Aug 2016 - Dec 2019*

- **FYP**: Interfacing and Testing of Localization Sensors on An Autonomous Vehicle(10356/136704)
- **Specification**: Robotics and Control (Honours Distinction)

Papers Accepted

- **ACMMM 2022 (CCF-A)**: Xingshuo Han, Guowen Xu, Yuan Zhou, **Xuehuan Yang**, Jiwei Li, Tianwei Zhang, Physical Backdoor Attacks to Lane Detection Systems in Autonomous Driving, ACM International Conference on Multimedia (MM), October 2022 (arxiv 2203.00858)
- **TIFS 2022 (CCF-A)**: Jianfei Sun, Guowen Xu, Tianwei Zhang, **Xuehuan Yang**, Mamoun Alazab, Robert Deng, Verifiable, Fair and Privacy-preserving Broadcast Authorization for Flexible Data Sharing in Clouds, Accepted by IEEE Transactions on Information Forensics & Security(TIFS)(9969631)
- **TIFS 2022 (CCF-A)**: Jianfei Sun, Guowen Xu, Tianwei Zhang, **Xuehuan Yang**, Mamoun Alazab, Robert Deng, Privacy-Aware and Security-Enhanced Efficient Matchmaking Encryption, Accepted by IEEE Transactions on Information Forensics & Security(TIFS)(10180078)
- **TDSC 2023 (CCF-A)**: Jianfei Sun, Guowen Xu, Hongwei Li, Tianwei Zhang, Cong Wu, **Xuehuan Yang**, Robert Deng, Sanitizable Cross-domain Access Control with Policy-driven Dynamic Authorization, Accepted by IEEE Transactions on Dependable and Secure Computing(TDSC)(10891832)

Award

- **NTU Award**: Graduated With Honours (Distinction)
- **Math Award**: 3rd Place, Chinese Mathematics Olympiad (CMO) competition hosted by the Chinese National High School Mathematics League
- **Chemical Award**: 3rd Place, Chinese Chemical Olympiad Competition hosted by Anhui Province
- **Province Award**: Merit Student in High School by Hefei City
- **CCA**: National Registry of Coaches (NROC) Coach by Sport Singapore && National International Inline Skating Coach Level 1 by Inline Certificate Program && Ministry of Education (MOE) Inline Skating Instructor by MOE Singapore && NTU Inline Skating Club Leader and Coach
- **Driving License**: International Driving License (Class 3, 4, and 5) by Singapore Safety Driving Centre. Operated and tested autonomous Toyota cars, vehicles, sweeper vehicles, and autonomous buses at NTU's Centre of Excellence for Testing & Research of Autonomous Vehicles (CETRAN). Driving electric prime movers at port and public road.